

Thank you for your interest in this work test. Please spend at most ten hours on the below prompts, and please email your completed memo and one script, in the language of your choosing, that produces the results discussed in your memo to christina@forecastingresearch.org. **A non-technical person like our CEO Josh Rosenberg should be able to read your memo and come away with an understanding of how different aggregation methods perform on this dataset and your ideas as to why.** If you want to do this exercise in a language other than Python or R, please check with Christina first.

You will be evaluated on your ability to understand and process the data provided; how well you communicate your process; and the quality of your code (you only have ten hours for this task, so it doesn't need to be the most efficient implementation, but it should be readable and tidy).

Please also note that:

1. Evaluators should be able to run your script by changing the working directory to one containing the raw data in the first line of your script.
2. Please include carefully-written comments in your code so that we can understand what each line/section of your script does, based only on reading the script itself.
3. Feel free to use libraries that assist in your workflow. Just ensure that any library you use can be installed with the current version of R using `install.packages()` or Python via `pip`. If you decide to use a library that's not as mainstream as e.g. `dplyr` or `numpy`, include a brief comment about what it's for.

Step 1: Download the raw forecasting data from the HFC forecasting tournament from: <https://dataverse.harvard.edu/dataverse/hfc>. Focusing on RCTA, the first year of competition, write a paragraph describing the structure of the three main raw data files.

Step 2: Take all individual forecasts, and aggregate the individual-level forecasts in the following five ways. The result should be five data points for each question-day pair, using the most recent forecast from each forecaster on that day (i.e. for each day that the question had any forecasts, each of the following aggregates):

- Method 1: Raw Mean
- Method 2: Median
- Method 3: Geometric Mean
- Method 4: Trimmed Mean (trimming top and bottom 10% of forecasts)
- Method 5: Geometric Mean of odds, as described here: <https://forum.effectivealtruism.org/posts/sMjcjnpoAQCcEdL2/when-pooling-forecasts-use-the-geometric-mean-of-odds>

Write 1-2 paragraphs describing any data cleaning you did to produce the aggregate forecasts described above. You may want to filter the forecasts in some way before aggregating. If you do any cleaning or filtering, explain your reasoning.

Step 3: Now, calculate how accurate the aggregate forecast from each aggregation method was, comparing the aggregated forecast to the resolved outcome of each question. Include a table with the results and a paragraph describing why certain methods were more accurate than others.

Step 4: Finally, propose and implement an improvement to the best of the five aggregation methods above. To calculate the aggregate score on a question-day level (eg. question Q on day T), you may **only** use information available through day T-1, including but not limited to (these are just suggestions):

1. Prior forecasting accuracy on resolved questions
2. Staleness of forecasts
3. Variance or other moments of forecasts within question
4. Forecaster activity

Why is your method of aggregation better than the five methods described in Step 3? Please explain.